

# Document storage in property management

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## User

Should a multi-tenant property management system store documents like leases meeting minutes and contractor invoices online for customers

## Assistant

Yes, a modern multi-tenant property management system (PMS) **should** store documents like leases, meeting minutes, and contractor invoices online, but only if architected and governed with strict attention to security, compliance, data isolation, and user experience.

Here's a structured breakdown to help you decide and implement it correctly:

### ✅ Why Online Storage is Recommended

| Benefit                         | Impact                                                                                                   |
|---------------------------------|----------------------------------------------------------------------------------------------------------|
| Accessibility & Transparency    | Tenants, owners, and managers can access documents from anywhere, reducing friction and support tickets. |
| Automated Workflows             | Enables e-signatures (ESIGN/UETA/eIDAS), approval routing, payment matching, and automated reminders.    |
| Version Control & Audit Trails  | Tracks edits, uploads, and access logs critical for legal disputes and financial audits.                 |
| Disaster Recovery & Scalability | Cloud object storage (S3, Azure Blob, GCS) offers built-in redundancy, backup, and elastic scaling.      |
| Cost & Compliance Efficiency    | Eliminates physical storage, reduces paper workflows, and simplifies retention/purging policies.         |

### ⚠️ Key Risks & Considerations

| Risk                       | Mitigation                                                                                                                                          |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Breach / PII Exposure | Leases & invoices contain sensitive personal/financial data. Use encryption (AES-256 at rest, TLS in transit), strict IAM, and regular pen testing. |

| Risk                                  | Mitigation                                                                                                                                                                |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Multi-Tenant Data Leakage</b>      | Never rely on UI-level controls. Enforce tenant isolation via <code>tenant_id</code> scoping, row-level security, and network policies. Audit access logs rigorously.     |
| <b>Legal Validity &amp; Retention</b> | Digital records must comply with jurisdictional laws (e.g., 3–7 years for leases, tax audit rules for invoices, GDPR/CCPA for PII). Automate retention & secure deletion. |
| <b>Vendor Lock-in / SLA Gaps</b>      | If using third-party storage, verify SOC 2 Type II, ISO 27001, data residency options, and clear data ownership/export clauses.                                           |
| <b>Customer Trust &amp; UX</b>        | Poor portal design, slow loads, or confusing permissions will drive users to offline workarounds. Prioritize intuitive, role-based access.                                |

## Best Practices for Implementation

### 1. Architecture

- Store files in **object storage** (not the relational DB). Keep metadata, permissions, and links in your database.
- Use **presigned URLs** or secure proxy endpoints for access. Never expose raw storage paths.
- Implement **tenant-scoped namespaces** (e.g., `s3://pms-bucket/tenant-{id}/property-{id}/`)

### 2. Access & Identity

- Role-based access control (RBAC): owner, tenant, property\_manager, admin, auditor
- Enforce MFA for staff/owners. Tenants get secure, expiring links or portal logins.
- Log all access, downloads, and edits immutably.

### 3. Compliance & Governance

- Automate document retention & secure deletion per jurisdiction.
- Support data residency (EU, CA, etc.) if operating internationally.
- Validate e-signature workflows against ESIGN/UETA (US) or eIDAS (EU).
- Provide customers with data export tools (CSV, PDF, ZIP) for portability.

### 4. Customer Portal Design

- Tenants see only their leases, payment receipts, and maintenance docs.
- Owners see financials, vendor invoices, and performance reports.
- Managers see cross-property operational docs with approval workflows.
- Never allow cross-tenant document discovery or sharing without explicit consent.

## 🚫 When to Reconsider or Modify

- Highly regulated markets (e.g., government housing, healthcare-adjacent properties) may require on-prem or hybrid storage.
- Legacy systems or clients with strict data sovereignty laws may demand local backups or air-gapped archives.
- Very large PDFs/scans should use lifecycle policies (hot → cool → cold storage) to control costs.

## ✅ Bottom Line

Online storage is not just feasible—it’s **expected** in modern PMS platforms. The differentiator isn’t *whether* you store documents online, but *how securely, compliantly, and transparently* you do it. Partner with legal/compliance experts early, implement defense-in-depth security, and design the customer portal around controlled access and ease of use. Done right, it becomes a competitive advantage, not a liability.

### User

How should the content be encrypted

### Assistant

Encrypting documents in a multi-tenant property management system requires a **defense-in-depth approach** that balances security, performance, cost, and tenant isolation. Below is a production-ready encryption strategy aligned with industry standards (NIST, CIS, SOC 2, GDPR/CCPA).

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## 🔒 Core Encryption Architecture

| Layer                 | Recommendation                                       | Why                                                                                                   |
|-----------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| In Transit            | TLS 1.2+ (enforced via HSTS)                         | Prevents MITM during upload/download. Use mutual TLS for internal service-to-service calls if needed. |
| At Rest (Files)       | <b>Envelope Encryption</b> via Cloud KMS             | Industry standard. Balances security, performance, and multi-tenant isolation.                        |
| At Rest (Metadata/DB) | TDE + column-level encryption for PII/financials     | Protects structured data without breaking app functionality.                                          |
| Key Management        | Dedicated KMS (AWS KMS, Azure Key Vault, GCP KMS, or | Centralized, auditable, compliant key lifecycle.                                                      |

| Layer | Recommendation   | Why |
|-------|------------------|-----|
|       | HashiCorp Vault) |     |

## Envelope Encryption Pattern (Recommended)

Never call KMS per file during read/write. Use **envelope encryption**:

1. **Generate a Data Encryption Key (DEK)** on-demand (e.g., AES-256-GCM).
2. **Encrypt the file** locally with the DEK.
3. **Encrypt the DEK** with a Tenant-Specific Key Encryption Key (KEK) stored in KMS.
4. Store: encrypted\_file, encrypted\_DEK, KEK\_id, tenant\_id, kms\_region.
5. On read: Decrypt DEK using KEK → decrypt file → stream to user.

### Benefits:

- KMS API calls minimized (only for DEK wrapping/unwrapping)
- Tenant isolation enforced at key policy level
- Supports key rotation without re-encrypting files
- Compliant with FIPS 140-2/3 & GDPR

## Multi-Tenant Key Isolation

| Technique                 | Implementation                                                                                   |
|---------------------------|--------------------------------------------------------------------------------------------------|
| <b>Tenant-Scoped KEKs</b> | Create one KMS key per tenant (or per business). Name: pms-tenant-{id}-kek                       |
| <b>KMS IAM Policies</b>   | Restrict kms:Decrypt/kms:Encrypt to specific service roles + tenant_id condition key             |
| <b>Storage Namespace</b>  | s3://bucket/tenant-{id}/property-{id}/ + KMS key policy = defense in depth                       |
| <b>Key Rotation</b>       | Automated every 90 days. KMS auto-rotates KEK; DEKs remain valid (envelope pattern handles this) |
| <b>Never Hardcode</b>     | Keys loaded at runtime via KMS SDK. Zero secrets in code, env vars, or config files.             |

## Search, Metadata & Indexing

You **cannot search encrypted file content directly**. Handle this separately:

- **Extract metadata** (title, date, type, parties, amounts) before encryption.
- Store metadata in DB/search index (Elasticsearch/OpenSearch) with **RBAC enforcement**.

- Hash sensitive PII (e.g., SSN, tax IDs) with **salted bcrypt/Argon2** for lookup.
  - If full-text search on documents is required, use **searchable encryption** (e.g., Microsoft SEAL, OpenSearch IRSA) or extract text via OCR/NLP before encryption.
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## Backup & Disaster Recovery

- Backups must be encrypted with the **same KMS keys** or equivalent regional keys.
  - Use **immutable storage** (S3 Object Lock, Azure Immutable Blob) for audit trails & compliance.
  - Cross-region backups: Replicate KMS keys or use **dual-auth KMS** with geographic restrictions.
  - Test restoration with key access quarterly. Document runbooks for key recovery.
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## Compliance & Auditing

| Requirement                  | Implementation                                                                                 |
|------------------------------|------------------------------------------------------------------------------------------------|
| <b>Audit Trails</b>          | Log all kms:Decrypt/kms:Encrypt calls with tenant_id, user_id, ip, timestamp. Forward to SIEM. |
| <b>Data Residency</b>        | Use regional KMS + storage. Enforce via IAM condition aws:RequestedRegion.                     |
| <b>GDPR/CCPA</b>             | Support data export & secure deletion. Automate retention policies. Log key destruction.       |
| <b>SOC 2 / ISO 27001</b>     | Maintain key inventory, rotation logs, access reviews, and penetration test reports.           |
| <b>Customer Transparency</b> | Provide tenants with access logs, key usage reports, and export tools upon request.            |

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## Implementation Checklist

- ☐ Deploy cloud-native KMS with tenant-scoped keys
- ☐ Implement envelope encryption library (e.g., AWS KMS SDK, Azure SDK, or open-source libsuid)
- ☐ Enforce TLS 1.2+ + HSTS + certificate pinning for mobile
- ☐ Configure KMS IAM policies with tenant\_id condition keys
- ☐ Set up automated key rotation (90 days) + backup key escrow
- ☐ Separate metadata index from encrypted file storage
- ☐ Log all KMS calls to SIEM + enable KMS audit trails
- ☐ Document backup/DR key access procedures
- ☐ Conduct third-party penetration test + key access review
- ☐ Provide tenant portal with access logs & secure export



## When to Consider Client-Side Encryption (CSE)

Use CSE only if:

- Handling government, healthcare, or highly regulated data
- Tenants require **zero-trust** where even your infrastructure can't decrypt
- Legal/compliance mandates customer holds the keys

⚠️ **Trade-offs:** Higher complexity, broken server-side search, key recovery challenges, performance overhead. Not recommended for standard residential/commercial PMS unless explicitly required.

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## Bottom Line

Encrypt with **envelope encryption + tenant-scoped KMS keys**, enforce isolation at the **IAM/KMS policy layer**, separate **metadata indexing** from file encryption, and automate **key rotation + audit logging**. This delivers bank-grade security without sacrificing performance, compliance, or multi-tenant safety. Partner with your cloud provider's security team early to validate KMS policies and backup key access paths.